32. We have $e = (a*b)*(a*b)$, and $(a*a)*(b*b) = e*e = e$ also. Thus $a*b*a*b = a*a*b*b$. Using left and right cancellation, we have $b*a = a*b$.

33. Let $P(n) = (a*b)^n = a^n * b^n$. Since $(a*b)^1 = a*b = a^1*b^1$, we see $P(1)$ is true. Suppose $P(k)$ is true. Then $(a*b)^{k+1} = (a*b)^k*(a*b) = (a^k*b^k)*(a*b) = [a^k*(b^k*a)]*b = [a^k*(a*b^k]*b = [(a^k*a)*b^k]*b = (a^{k+1}*b^k)*b = a^{k+1}*(b^k*b) = a^{k+1}*b^{k+1}$. This completes the induction argument.

34. The elements $e, a, a^2, a^3, \cdots, a^m$ aren't all different since G has only m elements. If one of $a, a^2, a^3, \cdots, a^m$ is e, then we are done. If not, then we must have $a^i = a^j$ where $i < j$. Repeated left cancellation of a yields $e = a^{j-i}$.

35. We have $(a*b)*(a*b) = (a*a)*(b*b)$, so $a*[b*(a*b)] = a*[a*(b*b)]$ and left cancellation yields $b*(a*b) = a*(b*b)$. Then $(b*a)*b = (a*b)*b$ and right cancellation yields $b*a = a*b$.

36. Let $a*b = b*a$. Then $(a*b)' = (b*a)' = a'*b'$ by Corollary 4.17. Conversely, if $(a*b)' = a'*b'$, then $b'*a' = a'*b'$. Then $(b'*a')' = (a'*b')'$ so $(a')'*(b')' = (b')'*(a')'$ and $a*b = b*a$.

37. We have $a*b*c = a*(b*c) = e$, which implies that $b*c$ is the inverse of a. Therefore $(b*c)*a = b*c*a = e$ also.

38. We need to show that a left identity element is a right identity element and that a left inverse is a right inverse. Note that $e*e = e$. Then $(x'*x)*e = x'*x$ so $(x')'*(x'*x)*e = (x')'*(x'*x)$. Using associativity, $[(x')'*x']*x*e = [(x')'*x']*x$. Thus $(e*x)*e = e*x$ so $x*e = x$ and e is a right identity element also. If $a'*a = e$, then $(a'*a)*a' = e*a' = a'$. Multiplication of $a'*a*a' = a'$ on the left by $(a')'$ and associativity yield $a*a' = e$, so a' is also a right inverse of a.

39. Using the hint, we show there is a left identity element and that each element has a left inverse. Let $a \in G$; we are given that G is nonempty. Let e be a solution of $y*a = a$. We show at $e*b = b$ for any $b \in G$. Let c be a solution of the equation $a*x = b$. Then $e*b = e*(a*c) = (e*a)*c = a*c = b$. Thus e is a left identity. Now for each $a \in G$, let a' be a solution of $y*a = e$. Then a' is a left inverse of a. By Exercise 38, G is a group.

40. It is easy to see that $\langle G, *\rangle$ is a group, because the order of multiplication in G is simply reversed: $(a*b)*c = a*(b*c)$ follows at once from $c\cdot(b\cdot a) = (c\cdot b)\cdot a$, the element e continues to act as identity element, and the inverse of each element is unchanged.

Let $\phi(a) = a'$ for $a \in G$, where a' is the inverse of a in the group $\langle G, \cdot\rangle$. Uniqueness of inverses and the fact that $(a')' = a$ show at once that ϕ is one to one and onto G. Also, $\phi(a\cdot b) = (a\cdot b)' = b'\cdot a' = a'*b' = \phi(a)*\phi(b)$, showing that ϕ is an isomorphism of $\langle G, \cdot\rangle$ onto $\langle G, *\rangle$.

41. Let $a, b \in G$. If $g*a*g' = g*b*g'$, then $a = b$ by group cancellation, so i_g is a one-to-one map. Because $i_g(g'*a*g) = g*g'*a*g*g' = a$, we see that i_g maps G onto G. We have $i_g(a*b) = g*a*b*g' = g*a*(g'*g)*b*g' = (g*a*g')*(g*b*g') = i_g(a)*i_g(b)$, so i_g satisfies the homomorphism property also, and is thus an isomorphism.

5. Subgroups

1. Yes **2.** No, there is no identity element. **3.** Yes **4.** Yes **5.** Yes

6. No, the set is not closed under addition. **7.** $\mathbb{Q}^+$ and $\{\pi^n \mid n \in \mathbb{Z}\}$

8. No. If $\det(A) = \det(B) = 2$, then $\det(AB) = \det(A)\det(B) = 4$. The set is not closed under multiplication.

9. Yes **10.** Yes, see Exercise 17 of Section 4.

11. No. If $\det(A) = \det(B) = -1$, then $\det(AB) = \det(A)\det(B) = 1$. The set is not closed under multiplication.

12. Yes, see Exercise 17 of Section 4.

13. Yes. Suppose that $(A^T)A = I_n$ and $(B^T)B = I_n$. Then we have $(AB)^T AB = B^T(A^T A)B = B^T I_n B = B^T B = I_n$, so the set of these matrices is closed under multiplication. Since ${I_n}^T = I_n$ and $I_n I_n = I_n$, the set contains the identity. For each A in the set, the equation $(A^T)A = I_n$ shows that A has an inverse A^T. The equation $(A^T)^T A^T = AA^T = I_n$ shows that A^T is in the given set. Thus we have a subgroup.

14. a) No, $\tilde{F}$ is not closed under addition. b) Yes

15. a) Yes b) No, it is not even a subset of $\tilde{F}$.

16. a) No, it is not closed under addition. b) Yes

17. a) No, it is not closed under addition. b) Yes

18. a) No, it is not closed under addition. b) No, it is not closed under multiplication.

19. a) Yes b) No, the zero constant function is not in $\tilde{F}$.

20. $G_1 \le G_1,\ G_1 < G_4$ $G_2 < G_1,\ G_2 \le G_2,\ G_2 < G_4,\ G_2 < G_7,\ G_2 < G_8$ $G_3 \le G_3,\ G_3 < G_5$
$G_4 \le G_4$ $G_5 \le G_5$ $G_6 \le G_5,\ G_6 \le G_6$ $G_7 < G_1,\ G_7 < G_4,\ G_7 \le G_7$
$G_8 < G_1,\ G_8 < G_4,\ G_8 < G_7,\ G_8 \le G_8$ $G_9 < G_3,\ G_9 < G_5,\ G_9 \le G_9$

21. **a.** -50, -25, 0, 25, 50 **b.** 4, 2, 1, $\frac{1}{2}, \frac{1}{4}$ **c.** $1, \pi, \pi^2, \frac{1}{\pi}, \frac{1}{\pi^2}$ **22.** $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

23. All the matrices $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ for $n \in \mathbb{Z}$. **24.** All the matrices $\begin{bmatrix} 3^n & 0 \\ 0 & 2^n \end{bmatrix}$ for $n \in \mathbb{Z}$.

25. All matrices of the form $\begin{bmatrix} 4^n & 0 \\ 0 & 4^n \end{bmatrix}$ or $\begin{bmatrix} 0 & -2^{2n+1} \\ -2^{2n+1} & 0 \end{bmatrix}$ for $n \in \mathbb{Z}$.

26. G_1 is cyclic with generators 1 and -1. G_2 is not cyclic. G_3 is not cyclic.
G_4 is cyclic with generators 6 and -6. G_5 is cyclic with generators 6 and $\frac{1}{6}$. G_6 is not cyclic.

To get the answers for Exercises 27 - 35, the student computes the given element to succesive powers (or summands). The first power (number of summands) that gives the identity element is the order of the cyclic subgroup. After students have studied Section 9, you might want to come back here and show them the easy way to handle the row permutations of the identity matrix in Exercises 33 - 35 by writing the permutation as a product of disjoint cycles. For example, in Exercise 35, row 1 is in row 4 place, row 4 is in row 2 place, and row 2 is in row 1 place, corresponding to the cycle (1,4,2). Row 3 is left fixed.

27. 4 **28.** 2 **29.** 3 **30.** 5 **31.** 4 **32.** 8 **33.** 2 **34.** 4 **35.** 3

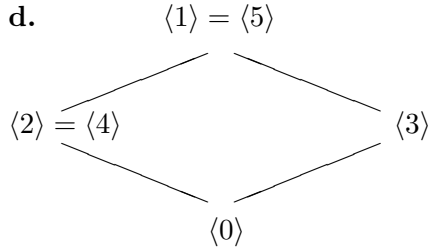
36. a.

$+_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

b. $\langle 0\rangle = \{0\}$
$\langle 2\rangle = \langle 4\rangle = \{0, 2, 4\}$
$\langle 3\rangle = \{0, 3\}$
$\langle 1\rangle = \langle 5\rangle = \mathbb{Z}_6$

c. 1 and 5

d.



37. Incorrect, the closure condition must be stated.

A **subgroup** of a group G is a subset H of G that is closed under the induced binary operation from G, contains the identity element e of G, and contains the inverse h^{-1} of each $h \in H$.

38. The definition is correct. **39.** T F T F F F F F T F

40. In the Klein 4-group, the equation $x^2 = e$ has all four elements of the group as solutions.

41. *Closure:* Let $a, b \in H$ so that $\phi(a), \phi(b) \in \phi[H]$. Now $(a * b) \in H$ because $H \leq G$. Since ϕ is an isomorphism, $\phi(a) *' \phi(b) = \phi(a * b) \in \phi[H]$, so $\phi[H]$ is closed under $*'$.

Identity: By Theorem 3.14, $e' = \phi(e) \in \phi[H]$.

Inverses: Let $a \in H$ so that $\phi(a) \in \phi[H]$. Then $a^{-1} \in H$ because H is a subgroup of G. We have $e' = \phi(e) = \phi(a^{-1} * a) = \phi(a^{-1}) *' \phi(a)$, so $\phi(a)^{-1} = \phi(a^{-1}) \in \phi[H]$.

42. Let a be a generator of G. We claim $\phi(a)$ is a generator of G'. Let $b' \in G'$. Because ϕ maps G onto G', there exists $b \in G$ such that $\phi(b) = b'$. Because a generates G, there exists $n \in \mathbb{Z}$ such that $b = a^n$. Because ϕ is an isomorphism, $b' = \phi(b) = \phi(a^n) = \phi(a)^n$. Thus G' is cyclic.

43. *Closure:* Let $S = \{hk \mid h \in H, k \in K\}$ and let $x, y \in S$. Then $x = hk$ and $y = h'k'$ for some $h, h' \in H$ and $k, k' \in K$. Because G is abelian, we have $xy = hkh'k' = (hh')(kk')$. Because H and K are subgroups, we have $hh' \in H$ and $kk' \in K$, so $xy \in S$ and S is closed under the induced operation.

Identity: Because H and K are subgroups, $e \in H$ and $e \in K$ so $e = ee \in S$.

Inverses: For $x = hk$, we have $h^{-1} \in H$ and $k^{-1} \in K$ because H and K are subgroups. Then $h^{-1}k^{-1} \in S$ and because G is abelian, $h^{-1}k^{-1} = k^{-1}h^{-1} = (hk)^{-1} = x^{-1}$, so the inverse of x is in S. Hence S is a subgroup.

44. If H is empty, then there is no $a \in H$.

45. Let H be a subgroup of G. Then for $a, b \in H$, we have $b^{-1} \in H$ and ab^{-1} H because H must be closed under the induced operation.

Conversely, suppose that H is nonempty and $ab^{-1} \in H$ for all $a, b \in H$. Let $a \in H$. Then taking $b = a$, we see that $aa^{-1} = e$ is in H. Taking $a = e$, and $b = a$, we see that $ea^{-1} = a^{-1} \in H$. Thus H contains the identity element and the inverse of each element. For closure, note that for $a, b \in H$, we also have $a, b^{-1} \in H$ and thus $a(b^{-1})^{-1} = ab \in H$.

46. Let $B = \{e, a, a^2, a^3, \cdots, a^{n-1}\}$ be a cyclic group of n elements. Then $a^{-1} = a^{n-1}$ also generates G, because $(a^{-1})^i = (a^i)^{-1} = a^{n-i}$ for $i = 1, 2, \cdots, n-1$. Thus if G has only one generator, we must have $n - 1 = 1$ and $n = 2$. Of course, $G = \{e\}$ is also cyclic with one generator.

47. *Closure:* Let $a, b \in H$. Because G is abelian, $(ab)^2 = a^2b^2 = ee = e$ so $ab \in H$ and H is closed under the induced operation.

Identity: Because $ee = e$, we see $e \in H$.

Inverses: Because $aa = e$, we see that each element of H is its own inverse. Thus H is a subgroup.

48. *Closure:* Let $a, b \in H$. Because G is abelian, $(ab)^n = a^nb^n = ee = e$ so $ab \in H$ and H is closed under the induced operation.

Identity: Because $e^n = e$, we see that $e \in H$.

Inverses: Let $a \in H$. Because $a^n = e$, we see that the inverse of a is a^{n-1} which is in H because H is closed under the induced operation. Thus H is a subgroup of G.

49. Let G have m elements. Then the elements $a, a^2, a^3, \cdots, a^{m+1}$ cannot all be different, so $a^i = a^j$ for some $i < j$. Then multiplication by a^{-i} shows that $e = a^{j-i}$, and we can take $j - i$ as the desired n.

50. Let $a \in H$ and let H have n elements. Then the elements $a, a^2, a^3, \cdots, a^{n+1}$ are all in H (because H is closed under the operation) and cannot all be different, so $a^i = a^j$ for some $i < j$. Then multiplication by a^{-i} shows that $e = a^{j-i}$ so $e \in H$. Also, $a^{-1} \in H$ because $a^{-1} = a^{j-i-1}$. This shows that H is a subgroup of G.

51. *Closure:* Let $x, y \in H_a$. Then $xa = ax$ and $ya = ay$. We then have $(xy)a = x(ya) = x(ay) = (xa)y = (ax)y = a(xy)$, so $xy \in H_a$ and H_a is closed under the operation.

Identity: Because $ea = ae = a$, we see that $e \in H_a$.

Inverses: From $xa = ax$, we obtain $xax^{-1} = a$ and then $ax^{-1} = x^{-1}a$, showing that $x^{-1} \in H_a$, which is thus a subgroup.

52. **a.** *Closure:* Let $x, y \in H_S$. Then $xs = sx$ and $ys = sy$ for all $s \in S$. We then have $(xy)s = x(ys) = x(sy) = (xs)y = (sx)y = s(xy)$ for all $s \in S$, so $xy \in H_S$ and H_S is closed under the operation.

Identity: Because $es = se = s$ for all $s \in S$, we see that $e \in H_S$.

Inverses: From $xs = sx$ for all $s \in S$, we obtain $xsx^{-1} = s$ and then $sx^{-1} = x^{-1}s$ for all $s \in S$, showing that $x^{-1} \in H_S$, which is thus a subgroup.

b. Let $a \in H_G$. Then $ag = ga$ for all $g \in G$; in particular, $ab = ba$ for all $b \in H_G$ because H_G is a subset of G. This shows that H_G is abelian.

53. *Reflexive:* Let $a \in G$. Then $aa^{-1} = e$ and $e \in H$ since H is a subgroup. Thus $a \sim a$.

Symmetric: Let $a, b \in G$ and $a \sim b$, so that $ab^{-1} \in H$. Since H is a subgroup, we have $(ab^{-1})^{-1} = ba^{-1} \in H$, so $b \sim a$.

Transitive: Let $a, b, c \in G$ and $a \sim b$ and $b \sim c$. Then $ab^{-1} \in H$ and $bc^{-1} \in H$ so $(ab^{-1})(bc^{-1}) = ac^{-1} \in H$, and $a \sim c$.

54. *Closure:* Let $a, b \in H \cap K$. Then $a, b \in H$ and $a, b \in K$. Because H and K are subgroups, we have $ab \in H$ and $ab \in K$, so $ab \in H \cap K$.

Identity: Because H and K are subgroups, we have $e \in H$ and $e \in K$ so $e \in H \cap K$.

Inverses: Let $a \in H \cap K$ so $a \in H$ and $a \in K$. Because H and K are subgroups, we have $a^{-1} \in H$ and $a^{-1} \in K$, so $a^{-1} \in H \cap K$.